

— NETWORK & SECURITY ENGINEER

Hassan Ismail.

CS & AI Student, South Valley National University

Working across two tracks: competitive programming in C++ (ECPC/ICPC) and network security.

"Failure is a milestone for success."

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LinkedIn Profile

Qena, Egypt

SELECTED WORK

- 01 **HA Cluster <-> SD-WAN Site-to-Site VPN**
Two FortiGate sites joined over an encrypted tunnel
- 02 **DMZ Segmentation**
Three-legged firewall isolating public-facing services
- 03 **Firewall Policy Engine**
Firewall rule-matching logic, rebuilt from scratch in C++



ID: H. ISMAIL

● AVAILABLE

What's actually behind the work

Trained hands-on through NTI's FortiOS Administrator program and independent practice.

CORE SKILLS

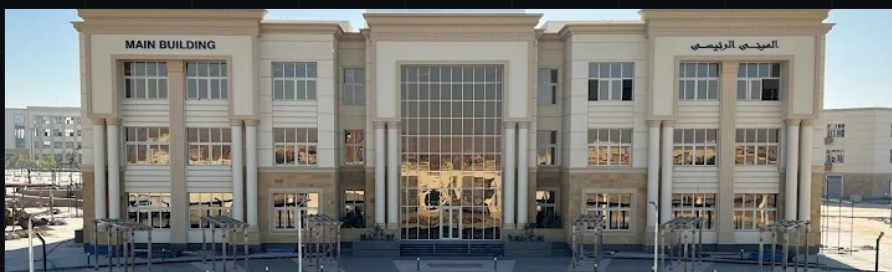
- Firewall Policy Design
- High Availability (HA) Clustering
- Site-to-Site IPsec VPN
- SD-WAN (Multi-WAN)
- Network Segmentation (DMZ)
- Static & Failsafe Routing
- CLI-Based Verification & Diagnostics

TOOLS & PLATFORMS

- FortiOS / FortiGate VM64
- FortiGate CLI Console
- Python (Intro + Intermediate)
- Network Fundamentals & Protocols

Currently building the fundamentals

Second-year Computer Science & AI student, with applied network-security practice and two completed Python courses alongside coursework.



ENTERING YEAR 3

South Valley National University

Faculty of Computer Science & Artificial Intelligence - CS & IT

Completed the second year, now entering the third. Coursework in programming, algorithms, and computer networks runs alongside the independent FortiGate work in this book.



NTI - 2026

National Telecommunication Institute

FortiOS Administrator -- Network Security, 60-hour track

Official Fortinet Academy curriculum: firewall policies, the Security Fabric, authentication, and security profiles (IPS, antivirus, web filtering, app control), plus IPsec VPN, SD-WAN, and HA -- the practical basis for the FortiGate projects in this book, built toward the NSE4 FortiOS Administrator path (exam NSE4_FGT_AD-7.6). Also completed two Microsoft-backed Python courses under an MCIT digital-skills program.



Intro to Python
Sep 14-23, 2025



Python - Intermediate
Sep 25-Oct 2, 2025

C++ contest record

Alongside network security, I compete in C++ with South Valley National University's ECPC/ICPC team.

APR 2025 - INDIVIDUAL

**ICPC ACPC Kickoff
Online Individual Contest**
Honorable Mention



AUG 2025 - PERSONAL CERTIFICATE

**ECPC Qualifications
Collegiate Prog. Contest, Day 8**
94th Place



AUG 2025 - TEAM CERTIFICATE

**ECPC Qualifications
Collegiate Prog. Contest, Day 8**
94th Place



● HASSAN ISMAIL

● COMPETITIVE PROGRAMMING -- CONTINUED

AUG 2026 - PERSONAL CERTIFICATE

ECPC Qualifications
Collegiate Prog. Contest, Day 8
93rd Place



AUG 2026 - TEAM CERTIFICATE

ECPC Qualifications
Collegiate Prog. Contest, Day 8
93rd Place



Services

Scope drawn directly from the work in this book.

01

Firewall & Policy Design

Rule sets built around least privilege - explicit deny where it matters, scoped ACCEPT rules everywhere else, all under UTM inspection.

Project 01 - 02

02

High Availability

Active-passive FortiGate clustering with unit priority and synchronization, so a single hardware failure doesn't take the network down.

Project 01

03

Site-to-Site VPN

IPsec tunnels between independent sites, with routing that prefers the tunnel and falls back to a blackhole route if it drops.

Project 01

04

SD-WAN

Multi-WAN edges grouped into a single zone, so outbound traffic keeps moving even if one link degrades.

Project 01

05

Network Segmentation

Zone-based design - LAN, DMZ, WAN - with firewall policy controlling what can reach what.

Project 02

06

Verification & Testing

Every path is confirmed from the CLI - sourced pings, tunnel status, session counters - before it's called done.

Project 01 - 02

Firewall Policy Engine

Real network security work above; this is the logic underneath it, rebuilt from scratch in C++ -- not configured through a GUI, but implemented.

Loads ACL-style policy from a file and evaluates packets against source/destination subnet -- real CIDR bitmask matching, not string comparison -- plus protocol and port range. Rules are checked top to bottom, first match wins, with a default implicit deny if nothing matches. A trace mode walks every rule against every packet and reports exactly which condition caused each one to match or fail.

Language: C++

Matching: CIDR / Protocol / Port

Verification: Per-rule trace mode

```
hassan@firewall-engine: ~/firewall-policy-engine

$ ./firewall_engine

--- tracing Packet 0 ---
Rule 0 [allow-everything]: src mismatch: 192.168.20.0/24, dst mismatch: 192.168.20.20/24, protocol mismatch: TCP, port mismatch: 1000-2000
Rule 1 [block-dmz-to-lan]: matches
Matched: [block-dmz-to-lan] -> DENY

--- tracing Packet 1 ---
Rule 0 [allow-everything]: matches
Rule 1 [block-dmz-to-lan]: src mismatch: 192.168.10.0/28, dst mismatch: 192.168.10.0/28, protocol mismatch: UDP, port mismatch: 100-200
Matched: [allow-everything] -> ALLOW

--- tracing Packet 2 ---
Rule 0 [allow-everything]: src mismatch: 192.168.20.0/24, dst mismatch: 192.168.20.20/24, port mismatch: 1000-2000
Rule 1 [block-dmz-to-lan]: src mismatch: 192.168.10.0/28, dst mismatch: 192.168.10.0/28, protocol mismatch: UDP, port mismatch: 100-200
No matching rule (implicit deny)
```

— PROJECT 01

HA Cluster <-> SD-WAN Site-to-Site VPN

P1 runs two FortiGate VM64 units in an HA cluster (priority 250, both synchronized) fronting the 192.168.10.0/24 LAN. P2 runs a single FortiGate with two WAN links bonded into an SD-WAN virtual-wan-link, fronting the 192.168.20.0/24 LAN. A site-to-site IPsec tunnel connects the two, with static routes on both sides preferring the tunnel and falling back to a blackhole route so traffic never leaks out to the internet if the VPN drops.

Role: Primary / Secondary

Priority: 250

Tunnel: Up - both directions

Verification: CLI ping, 0% loss

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Administrators

Admin Profiles

Firmware

Fabric Management

Settings

HA

SNMP

Replacement Messages

FortiGuard

Feature Visibility

Certificates

Security Fabric

Log & Report

HA: Primary

admin

FortiGate VM64

FortiGate VM64

FortiGate-VM64-HA-backup (Primary)

FortiGate-VM64 (Secondary)

Refresh

Edit

Remove device from HA cluster

Status	Priority	Hostname	Serial No.	Role	System Uptime	Sessions	Throughput
Synchronized	250	FortiGate-VM64-H	FGVMEVJZDR22AM0A	Primary	1m 15s	42	78.00 kbps
Synchronized	250	FortiGate-VM64	FGVMEVJWGIS_7T31	Secondary	3m 32s	27	22.00 kbps

2

01 HA Status

Active-passive HA pair, both synchronized at priority 250 - live session count and throughput per unit.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection		10.255.1.2-10.255.1.254	2
Physical Interface 11							
port1	Physical Interface		172.21.111.113/255.255.240.0	PING HTTPS SSH HTTP FMG-Access			4
port2	Physical Interface		0.0.0.0/0.0.0.0				0
port3	Physical Interface		192.168.10.1/255.255.255.0	PING HTTPS SSH			3
port4	Physical Interface		0.0.0.0/0.0.0.0				0

0 Security Rating Issues

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02 Interfaces

port1 carries WAN/management (SSH, HTTPS, FMG-Access); port3 terminates the internal 192.168.10.0/24 LAN.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

HA: Primary

admin

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

Create New

Edit

Clone

Delete

Search

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0	Dynamic Gateway (172.21.96.1)	port1	Enabled	
vpn to p2_remote		vpn to p2	Enabled	VPN: vpn to p2 (Created by VPN wizard)
vpn to p2_remote		Blackhole	Enabled	VPN: vpn to p2 (Created by VPN wizard)

FORTINET

v7.0.15

3

03 Static Routing

Default route out port1, a primary route through the tunnel, and a blackhole route so nothing leaks if the VPN drops.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

Create New

Edit

Delete

Policy Lookup

Search

Export

Interface Pair View

By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
port3 → port1 1									
LAN to internet	all	all	always	ALL	ACCEPT		no-inspection	UTM	0 B
port3 → vpn to p2 1									
vpn to p2 → port3 1									
Implicit 1									

0 Security Rating Issues

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04 Firewall Policy

LAN-to-internet plus dedicated inbound/outbound rules for the VPN interface, all under UTM inspection.

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

FortiGate-VM64-HA-backup

Dashboard

Network

Policy & Objects

Security Profiles

VPN

Overlay Controller VPN

IPsec Tunnels

IPsec Wizard

IPsec Tunnel Template

SSL-VPN Portals

SSL-VPN Settings

SSL-VPN Clients

VPN Location Map

User & Authentication

System

Security Fabric

Log & Report

HA: Primary

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admin

+ Create New

Edit

Delete

Search

Tunnel	Interface Binding	Status	Ref.
Site to Site - FortiGate 1			
vpn to p2	port1	Up	4

FORTINET v7.0.15

0 Security Rating Issues

1

05 IPsec Tunnel

Site-to-site tunnel "vpn to p2," bound to port1, status Up.

● HASSAN ISMAIL

● SITE P1 -- HA CLUSTER

PROJECT 01

Role: Primary/Secondary

Priority: 250

LAN: 192.168.10.0/24

Tunnel: vpn to p2 - Up

The screenshot shows the FortiGate VM64-HA-backup CLI console. The left sidebar contains a navigation menu with options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report. The main console area displays the following commands and output:

```
FortiGate-VM64-HA-backup # execute ping-options source 192.168.10.1
FortiGate-VM64-HA-backup # execute ping 192.168.20.1
PING 192.168.20.1 (192.168.20.1): 56 data bytes
64 bytes from 192.168.20.1: icmp_seq=0 ttl=255 time=2.4 ms
64 bytes from 192.168.20.1: icmp_seq=1 ttl=255 time=1.2 ms
64 bytes from 192.168.20.1: icmp_seq=2 ttl=255 time=1.7 ms
64 bytes from 192.168.20.1: icmp_seq=3 ttl=255 time=1.5 ms
64 bytes from 192.168.20.1: icmp_seq=4 ttl=255 time=1.6 ms

--- 192.168.20.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.2/1.6/2.4 ms
FortiGate-VM64-HA-backup #
```

The bottom status bar shows the Fortinet logo, version v7.0.15, and a security rating of 0. The top right corner indicates the user is 'admin'.

06 CLI Verification

Sourced ping from the P1 LAN to the P2 LAN (192.168.20.1) - 5/5 packets, 0% loss, 1.6 ms average.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection			1
Physical Interface 11							
port1	Physical Interface		172.21.109.73/255.255.240.0	PING HTTPS SSH HTTP FMG-Access			3
port2	Physical Interface		172.21.98.128/255.255.240.0	PING HTTPS SSH			1
port3	Physical Interface		192.168.20.1/255.255.255.0	PING HTTPS SSH			3
port4	Physical Interface		0.0.0.0/0.0.0.0				0

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01 Interfaces

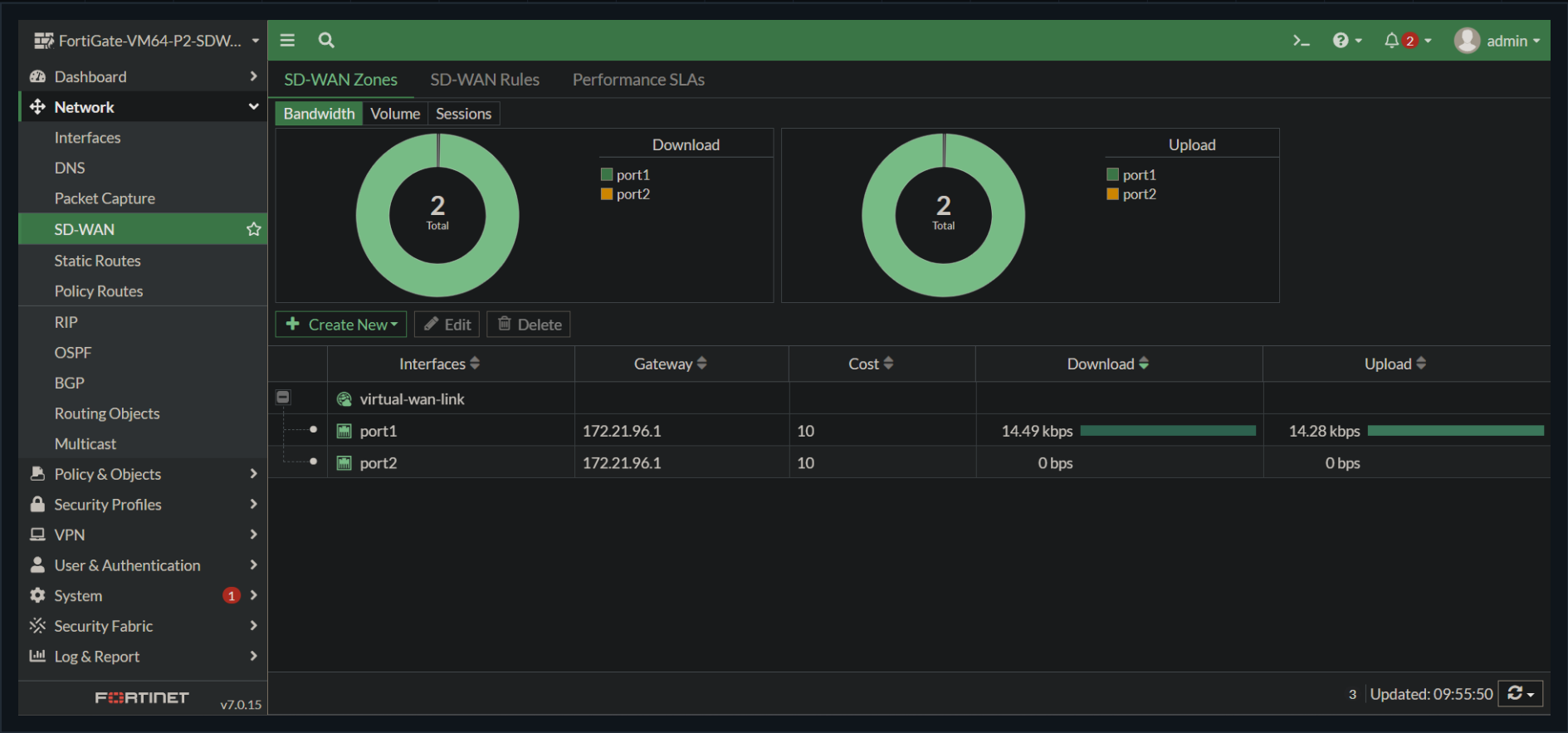
Two independent WAN links (port1, port2) plus the internal 192.168.20.0/24 LAN on port3.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up



02 SD-WAN Zone

Both WAN ports grouped into the virtual-wan-link at equal cost, with live download/upload throughput per member.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

- Interfaces
- DNS
- Packet Capture
- SD-WAN
- Static Routes
- Policy Routes
- RIP
- OSPF
- BGP
- Routing Objects
- Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

Create NewEditCloneDeleteSearch

Destination	Gateway IP	Interface	Status	Comments
0.0.0.0/0		virtual-wan-link	Enabled	
vpn to p1_remote		vpn to p1	Enabled	VPN: vpn to p1 (Created by VPN wizard)
vpn to p1_remote		Blackhole	Enabled	VPN: vpn to p1 (Created by VPN wizard)

FORTINET v7.0.15

3

03 Static Routing

Default route rides the SD-WAN zone; the route to P1 gets the same primary-plus-blackhole failsafe pattern as P1.

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

FortiGate-VM64-P2-SDW...

Dashboard

Network

Policy & Objects

Firewall Policy

IPv4 DoS Policy

Addresses

Internet Service Database

Services

Schedules

Virtual IPs

IP Pools

Protocol Options

Traffic Shaping

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

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🔔 2

admin

+ Create New

Edit

Delete

🔍 Policy Lookup

Search

🔍

📄 Export

Interface Pair View

By Sequence

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
🔍 port3 → virtual-wan-link 1									
lan to sdwan	all	all	always	ALL	✓ ACCEPT	Enabled	SSL no-inspection	UTM	0 B
+ port3 → vpn to p1 1									
+ vpn to p1 → port3 1									
+ Implicit 1									

0 Security Rating Issues

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04 Firewall Policy

LAN-to-SD-WAN policy for general egress, mirrored by inbound/outbound rules for the tunnel back to P1.

● HASSAN ISMAIL

- SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2

```
SD-WAN zone: virtual-wan-link
```

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

The screenshot displays the Fortinet FortiGate VM web interface. The left sidebar contains navigation menus for Dashboard, Network, Policy & Objects, Security Profiles, VPN, User & Authentication, System, Security Fabric, and Log & Report. The main area shows the 'IPsec Tunnels' configuration page under the 'VPN' section. A table lists existing tunnels, including one named 'vpn to p1' bound to 'port1'. The bottom status bar indicates 'v7.0.15' and '0 Security Rating Issues'.

Tunnel	Interface Binding	Status	Ref.
Site to Site - FortiGate			
vpn to p1	port1	Up	4

05 IPsec Tunnel

Site-to-site tunnel "vpn to p1," bound to port1, status Up.

● HASSAN ISMAIL

● SITE P2 -- SD-WAN

PROJECT 01

WAN links: port1 + port2

SD-WAN zone: virtual-wan-link

LAN: 192.168.20.0/24

Tunnel: vpn to p1 - Up

The screenshot shows the FortiGate VM64-P2-SDWAN CLI console. The left sidebar contains a navigation menu with options: Dashboard, Network, Policy & Objects, Security Profiles, VPN, Overlay Controller, IPsec Tunnels, IPsec Wizard, IPsec Tunnel Template, SSL-VPN Portals, SSL-VPN Settings, SSL-VPN Clients, VPN Location Map, User & Authentication, System, Security Fabric, and Log & Report. The main console area displays the following output:

```
FortiGate-VM64-P2-SDWAN # execute ping-options source 192.168.20.1

FortiGate-VM64-P2-SDWAN # execute ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
64 bytes from 192.168.10.1: icmp_seq=0 ttl=255 time=1.5 ms
64 bytes from 192.168.10.1: icmp_seq=1 ttl=255 time=0.9 ms
64 bytes from 192.168.10.1: icmp_seq=2 ttl=255 time=1.5 ms
64 bytes from 192.168.10.1: icmp_seq=3 ttl=255 time=0.9 ms
64 bytes from 192.168.10.1: icmp_seq=4 ttl=255 time=1.2 ms

--- 192.168.10.1 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.9/1.2/1.5 ms

FortiGate-VM64-P2-SDWAN #
```

The bottom of the console shows the FortiGate logo, version 7.0.15, and a security rating of 0. The user 'admin' is logged in.

06 CLI Verification

Sourced ping from the P2 LAN to the P1 LAN (192.168.10.1) - 5/5 packets, 0% loss, 1.2 ms average, confirming bidirectional reachability.

— PROJECT 02

DMZ Segmentation

A single FortiGate split into three security zones, so a compromised public-facing host in the DMZ can never pivot into the internal LAN. port1 faces the WAN, port2 hosts the internal LAN (192.168.10.0/24), and port3 is a dedicated DMZ (192.168.20.0/24) for exposed services. Policy does the isolating: DMZ-to-LAN traffic is explicitly denied, LAN-to-DMZ and LAN-to-WAN are permitted, and the implicit deny rule closes everything not named.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

LAN -> WAN: Accepted

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

FortiGate-VM64

Dashboard

Network

Interfaces

DNS

Packet Capture

SD-WAN

Static Routes

Policy Routes

RIP

OSPF

BGP

Routing Objects

Multicast

Policy & Objects

Security Profiles

VPN

User & Authentication

System

Security Fabric

Log & Report

FortiGate VM64

1 3 5 7 9 11 13 15 17 19 21 23

2 4 6 8 10 12 14 16 18 20 22 24

Create New Edit Delete Integrate Interface

Search

Group By Type

Name	Type	Members	IP/Netmask	Administrative Access	DHCP Clients	DHCP Ranges	Ref.
802.3ad Aggregate 1							
fortilink	802.3ad Aggregate		Dedicated to FortiSwitch	PING Security Fabric Connection			1
Physical Interface 10							
DMZ zone (port3)	Physical Interface		192.168.20.1/255.255.255.0	PING HTTPS SSH			3
LAN network (port2)	Physical Interface		192.168.10.1/255.255.255.0	PING HTTPS SSH			4
port1	Physical Interface		172.28.141.165/255.255.240.0	PING HTTPS SSH HTTP FMG-Access			1

0 SecurityRating Issues

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01 Interfaces

Dedicated DMZ (port3), internal LAN (port2), and WAN (port1) on a single FortiGate - each its own subnet.

Zones: WAN / LAN / DMZ

DMZ -> LAN: Denied

LAN -> DMZ: Accepted

FortiGate-VM64

- Dashboard
- Network
- Policy & Objects**
 - Firewall Policy**
 - IPv4 DoS Policy
 - Addresses
 - Internet Service Database
 - Services
 - Schedules
 - Virtual IPs
 - IP Pools
 - Protocol Options
 - Traffic Shaping
- Security Profiles
- VPN
- User & Authentication
- System
- Security Fabric
- Log & Report

Firewall Policy Configuration

Name	Source	Destination	Schedule	Service	Action	NAT	Security Profiles	Log	Bytes
DMZ zone (port3) → LAN network (port2)									
block dmz to lan	dmz network	lan network	always	ALL	DENY			All	0 B
LAN network (port2) → DMZ zone (port3)									
lan to dmz access	lan network	dmz network	always	ALL	ACCEPT	Disabled	SSL no-inspection	All	0 B
LAN network (port2) → port1									
lan to wan	lan network	all	always	ALL	ACCEPT	Enabled	SSL no-inspection	All	0 B
Implicit									
Implicit Deny	all	all	always	ALL	DENY			Enabled	0 B

0 Security Rating Issues | Updated: 09:07:56

02 Firewall Policy

DMZ-to-LAN explicitly denied, LAN-to-DMZ and LAN-to-WAN permitted, implicit deny closes everything else.

● HASSAN ISMAIL

— GET IN TOUCH

Open to network & security engineering work.

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GitHub Profile